

- None for healthy individuals
- Hydroxyzine may cause falsely elevated urinary concentrations of 17-hydroxycorticosteroids in certain lab tests (e.g., Porter-Silber reaction, Glenn-Nelson method)

Side Effects

How Drug Causes Side Effects

- Blocking histamine 1 receptors can cause sedation

Notable Side Effects

- Dry mouth, sedation, tremor



Life-Threatening or Dangerous Side Effects

- Rare convulsions (generally at high doses)
- Rare cardiac arrest, death (intramuscular formulation combined with CNS depressants)
- Bronchodilation
- Respiratory depression



Weight Gain

- Reported but not expected



Sedation

- Many experience and/or can be significant in amount
- Sedation is usually transient

What to Do About Side Effects

- Wait
- Wait
- Wait
- Switch to another agent

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- Anxiety: 50–100 mg 4 times a day
- Sedative: 50–100 mg oral, 25–100 mg intramuscular injection
- Pruritus: 75 mg/day divided into 3–4 doses

Dosage Forms

- Tablet 10 mg, 25 mg, 50 mg
- Capsule 25 mg, 50 mg, 100 mg

- Oral liquid 10 mg/5 mL
- Intramuscular 25 mg/mL, 50 mg/mL

How to Dose

- Oral dosing does not require titration
- Emergency intramuscular injection: initial 50–100 mg, repeat every 4–6 hours as needed
- Hydroxyzine intramuscular injection should not be given in the lower or mid-third of the arm and should only be given in the deltoid area if it is well developed
- In adults, hydroxyzine intramuscular injections may be given in the upper outer quadrant of the buttock or in the mid-lateral thigh

Dosing Tips

- Hydroxyzine may be administered intramuscularly initially, but should be changed to oral administration as soon as possible
- Tolerance usually develops to sedation, allowing higher dosing over time

Overdose

- Sedation, hypotension, possible QTc prolongation

Long-Term Use

- Evidence of efficacy for up to 16 weeks

Habit Forming

- No

How to Stop

- Taper generally not necessary

Pharmacokinetics

- Rapidly absorbed from gastrointestinal tract
- Mean elimination half-life approximately 20 hours



Drug Interactions

- If hydroxyzine is taken in conjunction with another CNS depressant, the dose of the CNS depressant should be reduced by half
- If hydroxyzine is used pre- or post-operatively, the dose of narcotic can be reduced
- If anticholinergic agents are used with hydroxyzine, the anticholinergic effects may be enhanced
- Hydroxyzine may reverse the vasopressor effect of epinephrine; patients requiring a vasopressor agent should use norepinephrine or metaraminol instead



Other Warnings/Precautions

- Hydroxyzine should not be administered subcutaneously, intra-arterially, or intravenously

Do Not Use

- If patient is in early stages of pregnancy
- If there is a proven allergy to hydroxyzine

Special Populations

Renal Impairment

- Dosage adjustment may not be necessary

Hepatic Impairment

- Dosage adjustment may not be necessary

Cardiac Impairment

- Hydroxyzine may be used to treat anxiety associated with cardiac impairment

Elderly

- Some patients may tolerate lower doses better
- Elderly patients may be more sensitive to sedative and anticholinergic effects
- Should be avoided in elderly patients with dementia

Children and Adolescents

- Anxiety, pruritus (6 and older): 50–100 mg/day in divided doses
- Anxiety, pruritus (under 6): 50 mg/day in divided doses
- Sedative: 0.6 mg/kg oral, 0.5 mg/lb intramuscular injection
- Small children should not receive hydroxyzine by intramuscular injection in the periphery of the upper quadrant of the buttock unless absolutely necessary because of risk of damage to the sciatic nerve
- Hyperactive children should be monitored for paradoxical effects

Pregnancy

- * Hydroxyzine is contraindicated in early pregnancy
- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Hydroxyzine intramuscular injection can be used prepartum, reducing narcotic requirements by up to 50%

Breast Feeding

- Unknown if hydroxyzine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk

- * Recommended either to discontinue drug or bottle feed

The Art of Psychopharmacology

Potential Advantages

- Has multiple formulations, including oral capsules, tablets, and liquid, as well as injectable
- No abuse liability, dependence, or withdrawal

Potential Disadvantages

- Patients with severe anxiety disorders
- Elderly patients
- Dementia patients

Primary Target Symptoms

- Anxiety
- Skeletal muscle tension
- Itching
- Nausea, vomiting



Pearls

- * A preferred anxiolytic for patients with dermatitis or skin symptoms such as pruritis

- Anxiolytic actions may be proportional to sedating actions
- Hydroxyzine tablets are made with 1,1,1- trichloroethane, which destroys ozone
- Hydroxyzine by intramuscular injection may be used to treat agitation during alcohol withdrawal
- Hydroxyzine may not be as effective as benzodiazepines or newer agents in the management of anxiety

Suggested Reading

Diehn F , Tefferi A . Pruritus in polycythaemia vera: prevalence, laboratory correlates and management . *Br J Haematol* 2001 ;**115** :619–21 .

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Ferreri M , Hantouche EG . Recent clinical trials of hydroxyzine in generalized anxiety disorder . *Acta Psychiatr Scand Suppl* 1998 ;**393** :102–8 .

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Guaiana G , Barbui C , Cipriani A. Hydroxyzine for generalised anxiety disorder . *Cochrane Database Syst Rev* 2010 ;**8** (12):CD006815 .

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Paton DM , Webster DR . Clinical pharmacokinetics of H1-receptor antagonists (the antihistamines) . *Clin Pharmacokinet* 1985 ;**10** :477–97 .

Iloperidone

Therapeutics

Brands

- Fanapt
- *see index for additional brand names*

Generic?

No

Class

- Neuroscience-based Nomenclature: dopamine and serotonin receptor antagonist (DS-RAn)
- Atypical antipsychotic (serotonin-dopamine antagonist; second-generation antipsychotic; also a mood stabilizer)

Commonly Prescribed for

(bold for FDA approved)

- **Schizophrenia**
- **Schizophrenia maintenance**
- Acute mania/mixed mania
- Other psychotic disorders

- Bipolar maintenance
- Bipolar depression
- Treatment-resistant depression
- Behavioral disturbances in dementia
- Behavioral disturbances in children and adolescents
- Disorders associated with problems with impulse control
- Posttraumatic stress disorder

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis and stabilizing affective symptoms
- Blocks serotonin 2A receptors, causing enhancement of dopamine release in certain brain regions and thus reducing motor side effects and possibly improving cognitive and affective symptoms
- Blockade of central alpha 1 adrenergic receptors may contribute to low potential for drug-induced parkinsonism

How Long Until It Works

- Psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior as well as on cognition
- Slow titration may delay antipsychotic effects during the first 1 to 2 weeks compared to some other antipsychotic drugs that do not require similar titration

- Classically recommended to wait at least 4–6 weeks to determine efficacy of drug, but in practice some patients may require up to 16–20 weeks to show a good response, especially on negative or cognitive symptoms

If It Works

- Most often reduces positive symptoms but does not eliminate them
- Can improve negative symptoms, as well as aggressive, cognitive, and affective symptoms in schizophrenia
- Most schizophrenia patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Perhaps 5–15% of schizophrenia patients can experience an overall improvement of greater than 50–60%, especially when receiving stable treatment for more than a year
- Such patients are considered super-responders or “awakeners” since they may be well enough to be employed, live independently, and sustain long-term relationships
- Continue treatment until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis
- For second and subsequent episodes of psychosis, treatment may need to be indefinite
- Even for first episodes of psychosis, it may be preferable to continue treatment indefinitely to avoid subsequent episodes

- Many bipolar patients may experience a reduction of symptoms by half or more
- Treatment may not only reduce mania but also prevent recurrences of mania in bipolar disorder

If It Doesn't Work

- Try one of the other atypical antipsychotics
- If 2 or more antipsychotic monotherapies do not work, consider clozapine
- Some patients may require treatment with a conventional antipsychotic
- If no first-line atypical antipsychotic is effective, consider higher doses or augmentation with valproate or lamotrigine
- Consider noncompliance and switch to another antipsychotic with fewer side effects or to an antipsychotic that can be given by depot injection (a depot formulation of iloperidone is in clinical testing)
- Consider initiating rehabilitation and psychotherapy such as cognitive remediation
- Consider presence of concomitant drug abuse



Best Augmenting Combos for Partial Response or Treatment Resistance

- Valproic acid (valproate, divalproex, divalproex ER)

- Other mood-stabilizing anticonvulsants (carbamazepine, oxcarbazepine, lamotrigine)
- Topiramate
- Lithium
- Benzodiazepines

Tests

Before Starting an Atypical Antipsychotic

- * Weigh all patients and track BMI during treatment
- Get baseline personal and family history of diabetes, obesity, dyslipidemia, hypertension, and cardiovascular disease
- * Get waist circumference (at umbilicus), blood pressure, fasting plasma glucose, and fasting lipid profile
- Determine if the patient
 - is overweight (BMI 25.0–29.9)
 - is obese (BMI >30)
 - has pre-diabetes (fasting plasma glucose 100–125 mg/dL)
 - has diabetes (fasting plasma glucose >126 mg/dL)
 - has hypertension (BP >140/90 mmHg)
 - has dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol)

- Treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

Monitoring After Starting an Atypical Antipsychotic

- * BMI monthly for 3 months, then quarterly
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics
- * Blood pressure, fasting plasma glucose, fasting lipids within 3 months and then annually, but earlier and more frequently for patients with diabetes or who have gained >5% of initial weight
- Treat or refer for treatment and consider switching to another atypical antipsychotic for patients who become overweight, obese, pre-diabetic, diabetic, hypertensive, or dyslipidemic while receiving an atypical antipsychotic
- * Even in patients without known diabetes, be vigilant for the rare but life-threatening onset of diabetic ketoacidosis, which always requires immediate treatment, by monitoring for the rapid onset of polyuria, polydipsia, weight loss, nausea, vomiting, dehydration, rapid respiration, weakness, and clouding of sensorium, even coma
- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and

iloperidone should be discontinued at the first sign of decline in WBC in the absence of other causative factors

- Patients at risk for electrolyte disturbances (e.g., patients on diuretic therapy) should have baseline and periodic serum potassium and magnesium measurements
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- Mechanism of weight gain and increased incidence of diabetes and dyslipidemia with atypical antipsychotics is unknown, and risks vary based on the particular agent

Notable Side Effects

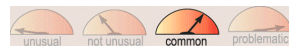
- * Orthostatic hypotension
- Sedation, dose-dependent dizziness, fatigue
- Dry mouth, nasal congestion
- Dose-dependent weight gain
- * May increase risk for diabetes and dyslipidemia
- Dose-dependent tachycardia
- Tardive dyskinesia (reduced risk compared to conventional antipsychotics)
- Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration



Life-Threatening or Dangerous Side Effects

- Hyperglycemia, in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, has been reported in patients taking atypical antipsychotics
- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis

- Rare seizures



Weight Gain

- Many experience and/or can be significant in amount
- May be less than for some antipsychotics, more than for others



Sedation

- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- Wait
- Wait
- Anticholinergics may reduce drug-induced parkinsonism when present
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain
- Switch to another atypical antipsychotic

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 12–24 mg/day in 2 divided doses

Dosage Forms

- Tablet 1 mg, 2 mg, 4 mg, 6 mg, 8 mg, 10 mg, 12 mg

How to Dose

- Initial 2 mg in 2 divided doses on day 1; 4 mg in 2 divided doses on day 2; 8 mg in 2 divided doses on day 3; 12 mg in 2 divided doses on day 4; 16 mg in 2 divided doses on day 5; 20 mg in 2 divided doses on day 6; 24 mg in 2 divided doses on day 7
- Maximum dose studied is 32 mg/day



Dosing Tips

- May titrate even slower in patients who develop side effects, especially orthostasis, or when adding or switching from another drug with alpha 1 antagonist properties
- Patients most vulnerable to side effects during titration would be those sensitive to orthostasis (e.g., the young, the elderly, those with cardiovascular problems, those taking concomitant vasoactive drugs)
- Slow dosing could lead to delayed onset of antipsychotic effects
- Once daily use seems theoretically possible because the half-life of iloperidone is 18–33 hours
- Some patients may respond to doses greater than 24 mg/day if tolerated
- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly
- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine
- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- Treatment should be suspended if absolute neutrophil count falls below 1000/mm³

- If treatment is discontinued for more than 3 days, it may need to be restarted following the initial titration schedule in order to maximize tolerability
- Iloperidone should be discontinued in patients with persistent QTc measurements of more than 500 msec

Overdose

- Sedation, tachycardia, hypotension

Long-Term Use

- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- Down-titration, especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)
- Rapid discontinuation could theoretically lead to rebound psychosis and worsening of symptoms

Pharmacokinetics

- Half-life 18–33 hours

- Metabolized by CYP450 2D6 and 3A4
- Food does not affect absorption



Drug Interactions

- May increase effects of antihypertensive agents
- May antagonize levodopa, dopamine agonists
- May enhance QTc prolongation of other drugs capable of prolonging QTc interval
- Inhibitors of CYP450 2D6 (e.g., paroxetine, fluoxetine, duloxetine, quinidine) may increase plasma levels of iloperidone and require a dosage reduction by one-half of iloperidone
- Inhibitors of CYP450 3A4 (e.g., nefazodone, fluvoxamine, fluoxetine, ketoconazole) may increase plasma levels of iloperidone and require a dosage reduction by one-half of iloperidone



Other Warnings/Precautions

- Use with caution in patients with conditions that predispose to hypotension (dehydration, overheating)
- Atypical antipsychotics have the potential for cognitive and motor impairment, especially related to sedation
- Atypical antipsychotics can cause body temperature dysregulation; use caution in patients who may experience conditions that increase body temperature (e.g., strenuous exercise, saunas, extreme heat, dehydration, or concomitant anticholinergic medication)

- Dysphagia has been associated with antipsychotic use, and iloperidone should be used cautiously in patients at risk for aspiration pneumonia
- Iloperidone prolongs QTc interval more than some other antipsychotics, an effect that is augmented by concomitant use of drugs that inhibit iloperidone metabolism
- Priapism has been reported with iloperidone
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy
- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If patient is taking agents capable of significantly prolonging QTc interval (e.g., pimozide, thioridazine, selected antiarrhythmics, moxifloxacin, sparfloxacin)
- If there is a history of QTc prolongation or cardiac arrhythmia, recent acute myocardial infarction, uncompensated heart failure
- If there is a proven allergy to iloperidone

Special Populations

Renal Impairment

- Dose adjustment not generally necessary

Hepatic Impairment

- Not recommended for patients with hepatic impairment

Cardiac Impairment

- Use in patients with cardiac impairment has not been studied, so use with caution because of risk of orthostatic hypotension
- Use with caution if patient is taking concomitant antihypertensive or alpha 1 antagonist
- Not recommended for patients with significant cardiovascular illness

Elderly

- Some patients may tolerate lower doses better
- Although atypical antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with atypical antipsychotics are at an increased risk of death compared to

placebo, and also have an increased risk of cerebrovascular events

- Consider pimavanserin for dementia-related psychosis or Parkinson's disease psychosis instead of an atypical antipsychotic

Children and Adolescents

- Safety and efficacy have not been established
- Children and adolescents using iloperidone may need to be monitored more often than adults and may tolerate lower doses better

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding

- When administered to pregnant rats during organogenesis, iloperidone was not teratogenic at doses up to 26 times the maximum recommended human dose (MRHD); however, at the highest dose it prolonged the duration of pregnancy and parturition; increased stillbirths, early intrauterine deaths, and incidence of developmental delays; and decreased postpartum pup survival
- When administered to pregnant rabbits during organogenesis, iloperidone was not teratogenic at doses up to 20 times the MRHD; however, at the highest dose, which was also a maternally toxic dose, it increased early intrauterine deaths and decreased fetal viability at term
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- Iloperidone may be preferable to anticonvulsant mood stabilizers if treatment is required during pregnancy
- National Pregnancy Registry for Atypical Antipsychotics: 1-866-961-2388,
<https://womensmentalhealth.org/research/pregnancyregistry/atypicalantipsychotic>

Breast Feeding

- Unknown if iloperidone is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed